

39.1

Activation Record (1)

39.1.1 Activation Record: ISRO CSE 2014 | Question: 51



Which of the following is NOT represented in a subroutine's activation record frame for a stack-based programming language?

- A. Values of local variables
- B. Return address
- C. Heap area
- D. Information needed to access non local variables

isro2014 programming activation-record

Answer key

39.2

Array (4)

39.2.1 Array: ISRO CSE 2008 | Question: 1



Which of the following is an illegal array definition?

- A. Type COLONGE : (LIME, PINE, MUSK, MENTHOL); var a : array [COLONGE] of REAL;
- B. var a : array [REAL] of REAL;
- C. var a : array ['A'...'Z'] of REAL;
- D. var a : array [BOOLEAN] of REAL;

isro2008 array

Answer key

39.2.2 Array: ISRO CSE 2009 | Question: 29



A one dimensional array A has indices 1....75. Each element is a string and takes up three memory words. The array is stored at location 1120 decimal. The starting address of A[49] is

- A. 1267
- B. 1164
- C. 1264
- D. 1169

isro2009 array programming

Answer key

39.2.3 Array: ISRO CSE 2020 | Question: 18



Consider a 2-dimensional array x with 10 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row-major format. If the first element $x[0][0]$ occupies the memory location with address 1000 and each element occupies only one memory location, which all locations (in decimal) will be holding a value of 10?

- a. 1018, 1019
- b. 1022, 1041
- c. 1013, 1014
- d. 1000, 1399

isro-2020 programming array normal

Answer key

39.2.4 Array: ISRO CSE 2020 | Question: 61



What is the output of the code given below?

```
#include<stdio.h>
int main()
{
    char name[]="satellites";
```

```

int len;
int size;
len= strlen(name);
size = sizeof(name);
printf("%d",len*size);
return 0;
}

```

A. 100

B. 110

C. 40

D. 44

isro-2020 programming array normal

Answer key

39.3

Array of Pointers (1)

39.3.1 Array of Pointers: ISRO CSE 2023 | Question: 68



Pointer p to an array of integers (of size 2 Bytes) is initialised to 200. What is the value of $p + 3$?

A. 206

B. 203

C. 212

D. 204

isro-cse-2023 array-of-pointers easy

Answer key

39.4

Functions (2)

39.4.1 Functions: ISRO CSE 2020 | Question: 78



In the following procedure

```

Integer procedure P(X,Y);
Integer X,Y;
value x;
begin
    K=5;
    L=8;
    P=X+y;
end

```

X is called by value and Y is called by name. If the procedure were invoked by the following program fragment

```

K=0;
L=0;
Z=P(K,L);

```

then the value of Z will be set equal to

A. 5

B. 8

C. 13

D. 0

isro-2020 programming functions normal

Answer key

39.4.2 Functions: ISRO2015-70



The following program

```

main()
{
    inc(); inc(); inc();
}
inc()

```

```
{
    static int x;
    printf("%d", ++x);
}
```

- A. prints 012
- B. prints 123
- C. prints 3 consecutive, but unpredictable numbers
- D. prints 111

isro2015 programming-in-c functions

Answer key

39.5

Identify Function (1)

39.5.1 Identify Function: ISRO-2013-63



Consider the following C code.

```
#include <stdio.h>
#include <math.h>
void main ()
{
    double pi = 3.1415926535;
    int a = 1;
    int i;
    for (i=0; i < 3; i++)
        if (a = cos(pi * i/2))
            printf("%d", 1);
        else printf("%d", 0);
}
```

What would the program print?

- A. 000
- B. 010
- C. 101
- D. 111

isro2013 programming-in-c identify-function output

Answer key

39.6

Loop (7)

39.6.1 Loop: ISRO CSE 2011 | Question: 63



What is the output of the following C code?

```
#include <stdio.h>

int main()
{
    int index;
    for(index=1; index<=5; index++)
    {
        printf("%d", index);
        if (index==3)
            continue;
    }
}
```

- A. 1245
- B. 12345
- C. 12245
- D. 12354

isro2011 programming-in-c loop

Answer key

39.6.2 Loop: ISRO CSE 2014 | Question: 38



How many lines of output does the following C code produce?

```
#include<stdio.h>
float i=2.0;
float j=1.0;
float sum = 0.0;
main()
{
    while (i/j > 0.001)
    {
        j+=j;
        sum=sum+(i/j);
        printf("%f\n", sum);
    }
}
```

- A. 8 B. 9 C. 10 D. 11

isro2014 programming-in-c loop output

Answer key

39.6.3 Loop: ISRO CSE 2017 | Question: 62



What will be the output of the following C code?

```
#include <stdio.h>
main()
{
    int i;
    for(i=0;i<5;i++)
    {
        int i=10;
        printf("%d" , i);
        i++;
    }
    return 0;
}
```

- A. 10 11 12 13 14 B. 10 10 10 10 10
C. 0 1 2 3 4 D. Compilation error

isro2017 programming-in-c loop output

Answer key

39.6.4 Loop: ISRO CSE 2017 | Question: 63



What does the following program do when the input is unsigned 16 bit integer?

```
#include<stdio.h>
main(){
    unsigned int num;
    int i;
    scanf("%u", &num);
    for(i=0;i<16;i++){
        printf("%d", (num<<i&1<<15)?1:0);
    }
}
```

- A. It prints all even bits from num B. It prints all odd bits from num
C. It prints binary equivalent of num D. None of above

isro2017 programming-in-c loop

Answer key

39.6.5 Loop: ISRO2015-71



Consider the following program fragment

```
i=6720; j=4;
while (i%j)==0
{
    i=i/j;
    j=j+1;
}
```

On termination j will have the value

- A. 4 B. 8 C. 9 D. 6720

isro2015 programming-in-c loop

Answer key

39.6.6 Loop: ISRO2015-73



The output of the following program is

```
main()
{
    static int x[] = {1,2,3,4,5,6,7,8}
    int i;
    for (i=2; i<6; ++i)
        x[x[i]]=x[i];
    for (i=0; i<8; ++i)
        printf("%d", x[i]);
}
```

- A. 1 2 3 3 5 5 7 8 B. 1 2 3 4 5 6 7 8
C. 8 7 6 5 4 3 2 1 D. 1 2 3 5 4 6 7 8

isro2015 programming-in-c loop output

Answer key

39.6.7 Loop: ISRO2015-75



The for loop

```
for (i=0; i<10; ++i)
printf("%d", i&1);
```

prints

- A. 0101010101 B. 0111111111 C. 0000000000 D. 1111111111

isro2015 programming-in-c loop

Answer key

39.7 Loop Invariants (1)

39.7.1 Loop Invariants: ISRO CSE 2007 | Question: 15



Consider the following pseudo-code

```
x:=1;
i:=1;
while (x <= 1000)
begin
    x:=2^x;
    i:=i+1;
end;
```

What is the value of i at the end of the pseudo-code?

A. 4

B. 5

C. 6

D. 7

isro2007 loop-invariants

Answer key

39.8

Macros (2)

39.8.1 Macros: ISRO CSE 2014 | Question: 3



What is the output of the following C program?

```
#include<stdio.h>
#define SQR(x) (x*x)

int main()
{
    int a;
    int b=4;
    a=SQR(b+2);
    printf("%d\n",a);
    return 0;
}
```

A. 14

B. 36

C. 18

D. 20

programming-in-c macros isro2014

Answer key

39.8.2 Macros: ISRO2015-76



Consider the following statements

```
#define hypotenuse (a, b) sqrt (a*a+b*b);
```

The macro call hypotenuse(a+2,b+3);

- A. Finds the hypotenuse of a triangle with sides $a + 2$ and $b + 3$
- B. Finds the square root of $(a + 2)^2$ and $(b + 3)^2$
- C. Is invalid
- D. Find the square root of $3 * a + 4 * b + 5$

programming-in-c macros isro2015

Answer key

39.9

Output (5)

39.9.1 Output: ISRO CSE 2014 | Question: 32



What is the output of the following C program?

```
#include<stdio.h>
void main(void){
    int shifty;
    shifty=0570;
    shifty=shifty>>4;
    shifty=shifty<<6;
    printf("The value of shifty is %o \n",shifty);
}
```

- A. The value of shifty is 15c0
- C. The value of shifty is 5700

- B. The value of shifty is 4300
- D. The value of shifty is 2700

programming-in-c isro2014 output

Answer key

39.9.2 Output: ISRO CSE 2017 | Question: 64



What is the output of the following program?

```
#include<stdio.h>
int tmp=20;
main()
{
    printf("%d", tmp);
    func();
    printf("%d", tmp);
}
func()
{
    static int tmp=10;
    printf("%d", tmp);
}
```

- A. 20 10 10 B. 20 10 20 C. 20 20 20 D. 10 10 10

isro2017 programming-in-c functions output

Answer key

39.9.3 Output: ISRO CSE 2018 | Question: 1



Consider the following program

```
{
    int x=1;
    printf("%d",(*char(char*)&x));
}
```

Assuming required header files are included and if the machine in which this program is executed is little endian, then the output will be

- A. 0 B. 99999999 C. 1 D. unpredictable

isro2018 programming output

Answer key

39.9.4 Output: ISRO CSE 2018 | Question: 60



What is the output of the following program?

```
main(){
    int x=2, y=5;
    if(x<y) return (x=x+y);
    else printf("z1");
    printf("z2");
}
```

- A. z2 B. z1z2
C. Compilation error D. None of these

isro2018 programming output

Answer key

39.9.5 Output: ISRO CSE 2018 | Question: 8



A language with string manipulation facilities uses the following operations.

head(*s*)- returns the first character of the string *s*

tails(*s*)- returns all but the first character of the string *s*

concat(*s1*, *s2*)- concatenates string *s1* with *s2*.

The output of concat(head(*s*), head(tail(tail(*s*)))), where *s* is *acbc* is:

- a. ab b. ba c. ac d. aa

isro2018 programming output

Answer key

39.10

Parameter Passing (1)

39.10.1 Parameter Passing: GATE CSE 2006 | Question: 56, ISRO2009-58



Consider the following code written in a pass-by-reference language like FORTRAN and these statements about the code.

```
subroutine swap(ix,iy)
  it = ix
L1 : ix = iy
L2 : iy = it
end
ia = 3
ib = 8
call swap (ia, ib+5)
print *, ia, ib
end
```

S1: The compiler will generate code to allocate a temporary nameless cell, initialize it to 13, and pass the address of the cell to swap

S2: On execution the code will generate a runtime error on line L1

S3: On execution the code will generate a runtime error on line L2

S4: The program will print 13 and 8

S5: The program will print 13 and -2

Exactly the following set of statement(s) is correct:

- A. S1 and S2 B. S1 and S4 C. S3 D. S1 and S5

gatecse-2006 programming parameter-passing normal out-of-syllabus-now isro2009

Answer key

39.11

Pointers (7)

39.11.1 Pointers: GATE CSE 2005 | Question: 1, ISRO2017-55



What does the following C-statement declare?

```
int (*f) (int * );
```

- A. A function that takes an integer pointer as argument and returns an integer
B. A function that takes an integer as argument and returns an integer pointer
C. A pointer to a function that takes an integer pointer as argument and returns an integer
D. A function that takes an integer pointer as argument and returns a function pointer

gatecse-2005 programming programming-in-c pointers easy isro2017

Answer key

39.11.2 Pointers: ISRO CSE 2014 | Question: 13



Which of the following is true with respect to Reference?

- A. A reference can never be NULL
- B. A reference needs an explicit dereferencing mechanism
- C. A reference can be reassigned after it is established
- D. A reference and pointer are synonymous

programming isro2014 pointers

Answer key

39.11.3 Pointers: ISRO CSE 2016 | Question: 20



What is the output of this C code?

```
#include<stdio.h>
void main()
{
    int k=5;
    int *p=&k;
    int **m=&p;
    printf("%d %d %d",k,*p,**m);
}
```

- A. 5 5 5
- B. 5 5 junk
- C. 5 junk junk
- D. compile time error

programming-in-c pointers isro2016

Answer key

39.11.4 Pointers: ISRO CSE 2016 | Question: 25



What will be output of the following program? Assume that you are running this program in little-endian processor.

```
#include<stdio.h>
int main()
{
    short a=320;
    char *ptr;
    ptr=(char *)&a;
    printf("%d",*ptr);
    return 0;
}
```

- A. 1
- B. 320
- C. 64
- D. Compilation error

programming-in-c pointers isro2016 little-endian-big-endian

Answer key

39.11.5 Pointers: ISRO CSE 2020 | Question: 62



What is output of the following 'C' code assuming it runs on a byte addressed little endian machine?

```
#include<stdio.h>
int main()
{
    int x;
    char *ptr;
    x=622,100,101;
    printf("%d",(*(char *)&x)*(x%3));
}
```

```
    return 0;
}
```

A. 622

B. 311

C. 22

D. 110

isro-2020 programming programming-in-c normal pointers

Answer key

39.11.6 Pointers: ISRO CSE 2020 | Question: 63



What is the output in a 32 bit machine with 32 bit compiler?

```
#include<stdio.h>
rer(int **ptr2, int **ptr1)
{
    int *ii;
    ii=*ptr2;
    *ptr2=*ptr1;
    *ptr1=ii;
    **ptr1=**ptr2;
    **ptr2+=**ptr1;
}

void main(){
    int var1=5, var2=10;
    int *ptr1=&var1,*ptr2=&var2;
    rer(&ptr1,&ptr2);
    printf("%d %d",var2,var1);
}
```

A. 60,70

B. 50,50

C. 50,60

D. 60,50

isro-2020 programming programming-in-c normal pointers

Answer key

39.11.7 Pointers: ISRO2015-72



Consider the following declaration:

```
int a, *b=&a, **c=&b;
```

The following program fragment

```
a=4; **c=5;
```

A. does not change the value of *a*
C. assigns the value of *b* to *a*

B. assigns address of *c* to *a*
D. assigns 5 to *a*

pointers programming-in-c isro2015

Answer key

39.12

Programming In C (7)

39.12.1 Programming In C: ISRO CSE 2008 | Question: 80



In C, what is the effect of a negative number in a field width specifier?

A. the values are displayed right justified
C. the values are displayed left justified

B. the values are displayed centered
D. the values are displayed as negative numbers

isro2008 programming programming-in-c non-gatecse

Answer key

39.12.2 Programming In C: ISRO CSE 2014 | Question: 47



The following three 'C' language statements is equivalent to which single statement?

```
y=y+1;  
Z=x+y;  
X=x+1
```

- A. $z = x + y + 2;$ B. $z = (x++) + (++y);$
C. $z = (x++) + (y++);$ D. $z = (x++) + (++y) + 1;$

isro2014 programming-in-c

Answer key

39.12.3 Programming In C: ISRO CSE 2016 | Question: 79



Which one of the following is correct about the statements given below?

- I. All function calls are resolved at compile time in C lang
II. All function calls are resolved at compile time in C++ lang

- A. Only II is correct B. Both I and II are correct
C. Only I is correct D. Both I and II are incorrect

isro2016 programming-in-c

Answer key

39.12.4 Programming In C: ISRO CSE 2018 | Question: 18



Consider the following C code segment:

```
#include <stdio.h>  
main()  
{  
    int i, j, x;  
    scanf("%d", &x);  
    i=1; j=1;  
    while (i<10) {  
        j=j*i;  
        i= i+1;  
        if(i==x) break;  
    }  
}
```

For the program fragment above, which of the following statements about the variables i and j must be true after execution of this program? [!(exclamation) sign denotes factorial in the answer]

- a. $(j = (x - 1)!) \wedge (i \geq x)$
b. $(j = 9!) \wedge (j = 10)$
c. $((j = 10!) \wedge (i = 10)) \vee ((j = (x - 1)!) \wedge (i = x))$
d. $(j = 9!) \wedge (i \geq 10) \vee ((j = (x - 1)!) \wedge (i = x))$

isro2018 programming-in-c programming

Answer key

39.12.5 Programming In C: ISRO CSE 2018 | Question: 70



Consider the following C program:

```
main()  
{  
    float sum= 0.0, j=1.0,i=2.0;  
    while(i/j>0.001){
```

```

j=j+1;
sum=sum+i/j;
printf("%f/n", sum);
}
}

```

- A. 0 - 9 lines of output
 C. 20 - 29 lines of output
 B. 10 - 19 lines out output
 D. More than 29 lines of output

isro2018 programming-in-c programming

Answer key

39.12.6 Programming In C: ISRO2015-68



Consider the following program fragment

```
if(a > b) if(b > c) s1; else s2;
```

s2 will be executed if

- A. $a \leq b$
 C. $b \geq c$ and $a \leq b$
 B. $b > c$
 D. $a > b$ and $b \leq c$

isro2015 programming programming-in-c

Answer key

39.12.7 Programming In C: ISRO2015-74



Which of the following has the compilation error in C?

- A. `int n = 17;`
 C. `float f = (float)99.32;`
 B. `char c = 99;`
 D. `#include <stdio.h>`

programming-in-c isro2015

Answer key

39.13 Recursion (3)

39.13.1 Recursion: GATE CSE 2004 | Question: 31, ISRO2008-40



Consider the following C function:

```

int f(int n)
{
    static int i = 1;
    if(n >= 5) return n;
    n = n+i;
    i++;
    return f(n);
}

```

The value returned by $f(1)$ is:

- A. 5
 B. 6
 C. 7
 D. 8

gatecse-2004 programming programming-in-c recursion easy isro2008

Answer key

39.13.2 Recursion: ISRO CSE 2018 | Question: 69



Let P be a procedure that for some inputs calls itself (i.e. is recursive). If P is guaranteed to terminate, which of the following statement(s) must be true?

- I. P has a local variable
- II. P has an execution path where it does not call itself
- III. P either refers to a global variable or has at least one parameter

- a. I only b. II only c. III only d. II and III only

isro2018 programming recursion

Answer key 

39.13.3 Recursion: ISRO CSE 2020 | Question: 59



Consider the following recursive C function that takes two arguments

```
unsigned int rer(unsigned int n, unsigned int r){
    if(n>0)return(n%r + rer(n/r,r));
    else return 0;
}
```

What is the return value of the function rer when it is called as $rer(513,2)$?

- A. 9 B. 8 C. 5 D. 2

isro-2020 programming normal recursion

Answer key 

39.14 Runtime Environment (3)

39.14.1 Runtime Environment: ISRO CSE 2007 | Question: 57



Which of the following programming language(s) provides garbage collection automatically

- A. Lisp B. C++ C. Fortan D. C

isro2007 programming runtime-environment

Answer key 

39.14.2 Runtime Environment: ISRO CSE 2014 | Question: 39



If only one memory location is to be reserved for a class variable, no matter how many objects are instantiated, then the variable should be declared as

- A. extern B. static C. volatile D. const

programming isro2014 runtime-environment

Answer key 

39.14.3 Runtime Environment: ISRO CSE 2017 | Question: 70



We use malloc and calloc for:

- A. Dynamic memory allocation B. Static memory allocation
C. Both dynamic memory allocation and static memory allocation D. None of these

isro2017 programming-in-c runtime-environment

Answer key 

39.15 Semantic Analysis (1)

39.15.1 Semantic Analysis: ISRO CSE 2018 | Question: 10



Which of the following comparisons between static and dynamic type checking incorrect?

- a. Dynamic type checking slows down the execution
- b. Dynamic type checking offers more flexibility to the programmers
- c. In contrast to Static type checking, dynamic type checking may cause failure in runtime due to type errors
- d. Unlike static type checking dynamic type checking is done during compilation

isro2018 programming semantic-analysis

Answer key

39.16

Structure (2)

39.16.1 Structure: ISRO CSE 2018 | Question: 2



Consider the following declaration :

```
structaddr {
    char city[10];
    char street[30];
    int pin;
};
struct {
    char name[30];
    int gender;
    struct addr locate;
} person, *kd = &person;
```

Then $*(kd \rightarrow name + 2)$ can be used instead of:

- A. $person.name + 2$
- B. $kd \rightarrow (name + 2)$
- C. $*((*kd).name + 2)$
- D. either A) or B), not C)

isro2018 programming structure

Answer key

39.16.2 Structure: ISRO CSE 2020 | Question: 74



Following declaration of an array of struct, assumes size of byte, short, int and long are 1,2,3 and 4 respectively. Alignment rule stipulates that n – byte field must be located at an address divisible by n , the fields in the struct are not rearranged, padding is used to ensure alignment. All elements of array should be of same size.

```
Struct complx
Short s
Byte b
Long l
Int i
End Complx
Complx C[10]
```

Assuming C is located at an address divisible by 8, what is the total size of C , in bytes?

- A. 150
- B. 160
- C. 200
- D. 240

isro-2020 programming normal structure

Answer key

39.17

Undefined Behaviour (1)

39.17.1 Undefined Behaviour: ISRO2015-69



If n has 3, then the statement $a[++n]=n++$;

- A. assigns 3 to $a[5]$
- B. assigns 4 to $a[5]$
- C. assigns 4 to $a[4]$
- D. what is assigned is compiler dependent

isro2015 programming-in-c non-gatecse undefined-behaviour

Answer key

39.18

Union (1)

39.18.1 Union: GATE CSE 2000 | Question: 1.17, ISRO2015-79



Consider the following C declaration:

```
struct {  
    short x[5];  
    union {  
        float y;  
        long z;  
    } u;  
}t;
```

Assume that the objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t , ignoring alignment consideration, is:

- A. 22 bytes
- B. 14 bytes
- C. 18 bytes
- D. 10 bytes

gatecse-2000 programming programming-in-c easy isro2015 union

Answer key

Answer Keys

39.1.1	C	39.2.1	B	39.2.2	C	39.2.3	X	39.2.4	B
39.3.1	A	39.4.1	B	39.4.2	B	39.5.1	C	39.6.1	B
39.6.2	D	39.6.3	B	39.6.4	C	39.6.5	C	39.6.6	A
39.6.7	A	39.7.1	B	39.8.1	A	39.8.2	D	39.9.1	D
39.9.2	B	39.9.3	X	39.9.4	D	39.9.5	A	39.10.1	B
39.11.1	C	39.11.2	A	39.11.3	A	39.11.4	C	39.11.5	D
39.11.6	D	39.11.7	D	39.12.1	C	39.12.2	B	39.12.3	D
39.12.4	D	39.12.5	D	39.12.6	D	39.12.7	X	39.13.1	C
39.13.2	D	39.13.3	D	39.14.1	A	39.14.2	B	39.14.3	A
39.15.1	D	39.16.1	C	39.16.2	X	39.17.1	D	39.18.1	C